Do Now · retrieval

Data capture, integrity and ACID transactions

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

How does DELETE differ from DROP TABLE?

Q2

What is a foreign key?

Q3

What does database consistency mean?

Q4

Array scores=[6,2,9,4]; replace element at index2 with element0+element1. Give final array and sum.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — Data capture gathers data, while validation, verification and referential integrity address different correctness concerns.

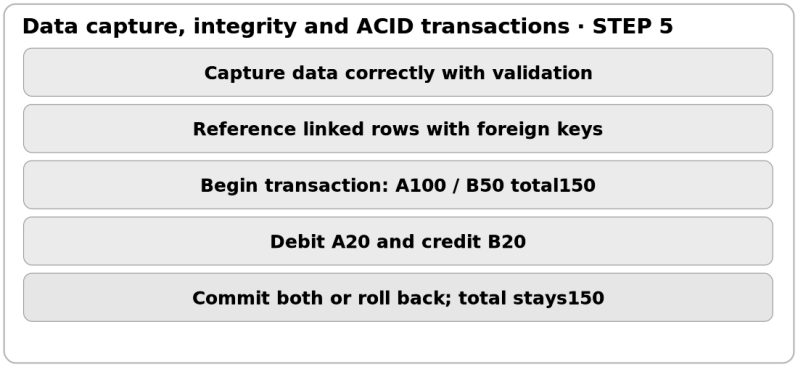
Core idea 2

Trace — Atomicity ensures a transaction is all-or-nothing; consistency preserves constraints; isolation addresses concurrent interference; durability keeps committed results.

Core idea 3

Explain — Row locking, transactions and appropriate constraints help prevent two concurrent changes from corrupting financial or stock records.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

Atomicity — All changes or none.

Consistency — Constraints preserved.

Isolation — Concurrent effects safely separated.

Durability — Committed changes persist.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

A stock purchase needs to reduce Stock and create Order together. Why use a transaction?

Follow the worked method

1. Predict — Both writes must succeed together or be undone so stock does not change without a valid order.

2. Check the answer — Explain which rule makes this result correct.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict

Only one successful transaction should reserve the final seat, preventing double booking.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

State the four ACID properties. [4]

Question 2

A transfer starts A100, B50. Move £20 and explain rollback on failure. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Two clients try to book one remaining place. Explain how transaction isolation and locking protect integrity. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A failed two-step bank transfer may safely leave the debit saved without the credit.

Use key vocabulary from memory

1. Explain Atomicity in your own words.

2. Explain Consistency in your own words.

3. Explain Isolation in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.